

5 (a) $R \cos 14^\circ = W = 8500$

$$\Rightarrow R = \frac{8500}{\cos 14^\circ}$$

B1

$$\begin{aligned}\text{Horizontal component of } R &= R \sin 14^\circ = \frac{8500}{\cos 14^\circ} \sin 14^\circ \\ &= 2100 \text{ N}\end{aligned}$$

B1

(b) Horizontal component of R provides the centripetal force for circular motion.

$$\Rightarrow m \frac{v^2}{r} = 2100$$

$$\Rightarrow \frac{8500 \times v^2}{9.81 \times 150} = 2100 \quad \text{M1}$$

$$\Rightarrow v = 19 \text{ m s}^{-1} \quad \text{A1}$$

- (c) There is a greater centripetal force to be provided at higher velocity. Hence, there is a horizontal component of frictional force acting on the car besides the reaction force R , providing the centripetal force required for its circular motion M1
 The car will tend to slide up the slope. A1

6 (a) (i) -When an electron transits from a higher energy level to lower energy level R1